

Lecture 9 - Introduction

Lecture 09 - Correlation and Regression - Introduction

Purpose of the Lecture

This lecture introduces regression as a way to study relationships between variables. Earlier lectures focused mostly on one variable at a time: its distribution, average, variability, sampling uncertainty, or a claim about its mean. Lecture 09 asks a different kind of question. When two variables are observed together, does one help summarize, explain, or predict the other?

This question appears constantly in finance and business. A firm may ask whether investment is associated with revenue. A marketing team may ask whether advertising spending helps explain sales. A credit analyst may ask whether leverage helps predict default risk. In each case, the data contain relationships, and regression gives a disciplined way to describe them.

The lecture begins with models. A model is a simplified mathematical description of reality. In a deterministic model, the input fully determines the output. In a statistical model, the same input can lead to different outcomes because unobserved forces also matter. Regression belongs to this second category. It estimates an average pattern while leaving room for unexplained variation.

The first part of the lecture develops simple linear regression. Students learn how to identify the dependent and explanatory variables, interpret the intercept and slope, compute fitted values and residuals, and understand the least-squares fitting rule. The lecture then connects regression to correlation and the coefficient of determination (R^2), and emphasizes the distinction between association and causation.

The second part turns to prediction. A fitted line can be useful inside the range of observed data, but it becomes more fragile outside that range. This leads to interpolation, extrapolation, model complexity, underfitting, overfitting, and the bias-variance tradeoff.

A non-graded optional appendix applies the same regression slope idea to financial beta.

Learning Objectives

This lecture aims to:

1. Explain why statistical models include unexplained variation.
2. Distinguish deterministic, statistical, linear, and nonlinear models.
3. Identify dependent and explanatory variables.
4. Interpret the intercept, slope, fitted value, and residual.
5. Explain the least-squares fitting rule.
6. Interpret correlation, squared correlation (r^2), and the coefficient of determination (R^2) carefully.
7. Separate association from causation.

8. Estimate and read a simple regression in Python.
9. Distinguish interpolation from extrapolation.
10. Explain underfitting, overfitting, and the bias-variance tradeoff.
11. Connect covariance, correlation, and the regression slope.

Learning Outcomes

By the end of this session, students should be able to:

1. Write a simple regression equation and interpret its coefficients.
2. Compute fitted values and residuals from a proposed line.
3. Explain how least squares chooses a fitted line.
4. Interpret correlation as standardized linear association.
5. Explain why the coefficient of determination (R^2) measures sample fit, not causality.
6. Recognize extrapolation and explain why it is risky.
7. Describe underfitting and overfitting in words.
8. Compute a slope from covariance, variance, or correlation.
9. Report a regression result with uncertainty and context.

Lecture Roadmap

The lecture begins with the idea of a model. A deterministic model gives an exact output. A statistical model estimates an average pattern and includes a noise term for what the model does not explain.

The lecture then introduces simple linear regression. A fitted line gives predicted values. Residuals measure the observed gaps between the data and the line. Least squares chooses the line with the smallest total squared residuals.

Next, the lecture connects regression with correlation. Correlation (r) gives the direction and strength of a linear association. In simple regression with an intercept, the coefficient of determination (R^2) equals (r^2), the squared correlation between (X) and (Y). This makes (R^2) useful for measuring sample fit, but it does not make the relationship causal.

Python is then used to estimate the regression. The output reports coefficients, uncertainty measures, confidence intervals, and (R^2). The official Python exercises use `data_lecture9_exercises.xlsx`, with sheet `data1` for the main regression and sheets `data2`, `data3`, and `data4` for comparing fitted lines. The separate multiple-regression lecture demonstration uses `data_lecture9.xlsx`, sheet `multivariate`. The aim is not to memorize the table. The aim is to know which quantities answer which questions.

The second part studies prediction. Interpolation stays inside the observed range of the data. Extrapolation goes outside that range and relies on stronger assumptions. The lecture also introduces model complexity: simple models can underfit, flexible models can overfit, and good prediction usually requires a balance.

The supplementary Python practice includes an optional train/test split. This is not part of the examinable slide material. It is included only to show, in code, how to estimate a line on one part of the data and check its predictions on held-out observations.

The non-graded optional appendix applies regression to financial beta. Beta is the slope in a regression of asset excess return on market excess return:

undefinedThis appendix is not part of the core or graded exercise set. It is included only as an additional finance application of the regression slope.

Central Idea

The central idea of the lecture is:

Regression estimates an average relationship between variables, and that relationship must be interpreted with uncertainty, context, and limits.

A fitted line can summarize a pattern and support prediction. It can also mislead when the model is used outside the data, when residual variation is ignored, or when association is treated as causation.

How to Work Through the Lecture

Start with the question. Decide what the output is and what input is being used to explain or predict it. Then read the fitted line: the intercept positions the line, and the slope gives the average change in the predicted output for a one-unit change in the input.

Next, focus on residuals. Residuals show what the fitted line leaves unexplained. They are not a side detail. They are the evidence that the model is only an approximation.

Then read the regression output slowly. The coefficient gives the estimate. The standard error and confidence interval describe uncertainty. (R^2) describes sample fit. None of these numbers removes the need for context.

Finally, keep the appendix separate from the core lecture. The official course objective is regression interpretation. Financial beta is an optional, non-graded application.

What to Pay Attention To

Three habits matter in this lecture.

First, define the question before defining the model. The same dataset can support different regressions.

Second, separate association from causation. Regression can be informative without being causal.

Third, keep prediction limits visible. Interpolation is usually safer than extrapolation.

End-of-Lecture Check

By the end of the lecture, you should be comfortable answering:

1. What is a statistical model?
2. What are the slope and intercept?
3. What is a fitted value?
4. What is a residual?
5. What does least squares minimize?
6. What does correlation measure?
7. What is the difference between squared correlation (r^2) and the coefficient of determination (R^2)?
8. Why does correlation not prove causation?
9. Why is extrapolation risky?

10. What is the difference between underfitting and overfitting?

11. How do covariance, correlation, and slope differ?